Irreducible Curriculum for Language Model Pretraining

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Abstract

Automatic data selection and curriculum design for training large language models is challenging, with only a few existing methods showing improvements over standard training. Furthermore, current schemes focus on domain-level selection, overlooking the more fine-grained contributions of each individual training point. It is difficult to apply traditional datapoint selection methods on large language models: most online batch selection methods perform two-times forward or backward passes, which introduces considerable extra costs with large-scale models. To mitigate these obstacles, we propose IRREDUCIBLE CURRICULUM as a curriculum learning algorithm for language model pretraining, which prioritizes samples with higher learnability. Specifically, to avoid prohibitive extra computation overhead, we simulate the sample loss along the main model's training trajectory using a small-scale proxy model. Our experiments on the RedPajama-1B dataset demonstrate a consistent improvement on validation perplexity across all 7 domains compared to random uniform baseline and the anti-curriculum strategy. Our method also reduces the sharpness of the network and illustrates a better 5-shot accuracy on MMLU benchmarks.

1 Introduction

The evolution of language models demonstrates impressive generation and reasoning abilities from a continuous scaling-up of model size and training corpus [Brown et al., 2020, Chowdhery et al., 2022, Kaplan et al., 2020, Hoffmann et al., 2022], along with very significant growth in computation cost. Besides the quantity [Kaplan et al., 2020] and quality [Lee et al., 2022, Longpre et al., 2023] of training data, several recent works show that the ordering [Chen et al., 2023] and composition [Xie et al., 2023] of data during the training can highly impact performance and efficiency. Xie et al. [2023] proposes DoReMi, which determines the optimal domain mixture to construct the pretraining corpus using auxiliary models; Chen et al. [2023] introduces an online selection scheme, which seeks to dynamically update the mixture of data from each skill-set at each training step. While DoReMi and Skill-it demonstrates great downstream performance, their curriculum are built upon the group-level where all the instances within one group (domain/skill) shares the same sampling probability, without looking inside each domains on the sample-specific attributions.

Nevertheless, few research have succeeded applying traditional sample-level selection schemes for large langauge model pretraining. Online batch selection methods [Loshchilov and Hutter, 2015, Katharopoulos and Fleuret, 2018, Jiang et al., 2019, Schaul et al., 2015] select hard samples with high loss or high gradient norm, which require a second forward/backward pass. That introduces large extra computation costs when the model size is large, which hurts the scalability. On the other side, Campos [2021] shows that the linguistic-based curriculum learning failed to improve on causal language model pretraining. Schaul et al. [2015] introduces Rho-Loss, which is an online batch selection scheme based on the gap between current training loss and an irreducible loss term

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measured using a proxy model. Inspired by the idea of proxy model approximation, we propose IRREDUCIBLE CURRICULUM as a curriculum learning algorithm, construct an ordered data stream by simulating the learning trajectory of large model through the lens of a small-scale proxy model. Specifically, we train the small-scale proxy model on a holdout set, and measure the loss on each data samples in the training set using the proxy model at different training stages. We estimate the *learnability score* of the training samples by the loss gap from the proxy model at early- and late-stage. The IRREDUCIBLE CURRICULUM would prioritize the samples with highest *learnability score* and proceed to the samples with lower *learnability score*. We present the details of our methodology in Section 3.

2 Background

2.1 Reducible Holdout Loss Selection (RHO-LOSS)

We recapitulate RHO-LOSS [Mindermann et al., 2022] with the emphasis on their data selection criterion and approximation using the proxy model. Considering a model $\mathcal{M}(\boldsymbol{\theta})$ parameterized by $\boldsymbol{\theta}$, and training corpus $\mathcal{D}_{train} = \{(x_i, y_i)\}_{i=1}^{n}$, online batch selection pre-samples a larger batch $B_t \in \mathcal{D}_{train}$ uniformly, from which it then selects a smaller subset $b_t \in B_t, |B_t| \gg |b_t|$.

At training step t, given the model has seen part of training corpus $\mathcal{D}_{\mathrm{t}}$, the output distribution from $\mathcal{M}(\boldsymbol{\theta})$ for an input x is denoted as $p(y|x; \mathcal{D}_{\mathrm{t}})$. Therefore, the log-likelihood objective on the holdout evaluation set after adding a new training point (x, y) can be written as $\log \mathrm{p}(y^{\mathrm{ho}}|x^{\mathrm{ho}}; \mathcal{D}_{\mathrm{t}} \cup (x, y))$. Applying Bayesian rule and conditional independence, the notion of the RHO-LOSS can be derived as,

$$
\begin{aligned}
& \log \mathrm{p}(y^{\mathrm{ho}}|x^{\mathrm{ho}}; \mathcal{D}_{\mathrm{t}} \cup (x, y)) \\
& = \log \frac{\mathrm{p}(y|x; x^{\mathrm{ho}}, y^{\mathrm{ho}}, \mathcal{D}_{\mathrm{t}})\, \mathrm{p}(y^{\mathrm{ho}}|x^{\mathrm{ho}}, x; \mathcal{D}_{\mathrm{t}})}{\mathrm{p}(y|x, x^{\mathrm{ho}}; \mathcal{D}_{\mathrm{t}})} \quad \text{Bayes rule} \\
& = \log \frac{\mathrm{p}(y|x; y^{\mathrm{ho}}, x^{\mathrm{ho}}, \mathcal{D}_{\mathrm{t}})\, \mathrm{p}(y^{\mathrm{ho}}|x^{\mathrm{ho}}; \mathcal{D}_{\mathrm{t}})}{\mathrm{p}(y|x; \mathcal{D}_{\mathrm{t}})} \quad \begin{array}{c}\text{conditional} \\ \text{independence}\end{array} \\
& \propto L\left[y|x; \boldsymbol{\theta}[t], \mathcal{D}_{\mathrm{t}}\right] - L\left[y|x; \boldsymbol{\theta}[t], \mathcal{D}_{\mathrm{ho}}, \mathcal{D}_{\mathrm{t}}\right] \\
& \approx L\left[y|x; \boldsymbol{\theta}[t], \mathcal{D}_{\mathrm{t}}\right] - L\left[y|x; \boldsymbol{\theta}[t], \mathcal{D}_{\mathrm{ho}}\right] \qquad (1) \\
& \approx \overbrace{\underbrace{L\left[y|x; \boldsymbol{\theta}[t], \mathcal{D}_{\mathrm{t}}\right]}_{\text{training loss}} - \underbrace{L\left[y|x; \boldsymbol{\theta}', \mathcal{D}_{\mathrm{ho}}\right]}_{\text{irreducible holdout loss (IL)}}}^{\text{reducible holdout loss}} \quad \begin{array}{c}\text{proxy-model} \\ \text{approximation}\end{array} \qquad (2)
\end{aligned}
$$

, where $\boldsymbol{\theta}'$ denotes a smaller scale proxy model pretrained on holdout set $\mathcal{D}_{\mathrm{ho}}$. To avoid retraining the model on $(\mathcal{D}_{\mathrm{t}} \cup \mathcal{D}_{\mathrm{ho}})$ at each step, the second term in Equ. (1) is approximated with the model trained on the holdout set $L\left[y|x; \boldsymbol{\theta}[t], \mathcal{D}_{\mathrm{ho}}, \mathcal{D}_{\mathrm{t}}\right] \approx L\left[y|x; \boldsymbol{\theta}[t], \mathcal{D}_{\mathrm{ho}}\right]$ *(irreducible holdout loss)*, such that the second term could be computed by a pretrained, reusable model. To further improve computational efficiency, the *small-scale proxy model* $\mathcal{M}_{proxy}(\boldsymbol{\theta}')$ is applied to compute the irreducible holdout loss. Mindermann et al. [2022] empirically proved that the approximation can preserve the desired properties with significantly lower computational costs.

Qualitatively, RHO-LOSS down-weighs the *redundant*, *noisy* samples and *outliers*: redundant points have low training loss and low irreducible holdout loss, while noisy samples and outliers have high training loss and high irreducible holdout loss, both lead to a low RHO-LOSS score. While the approximation alleviates the computational burden, the first term in Equ. (2) still requires one forward pass through the proxy model, and two forward passes through the large model at every step: On a pre-sampled batch B_t, the scoring takes one pass through the proxy model and the large model. Then the large model would be trained with a second pass on the top-ranked $|b_t|$ with highest scores. Given that it takes C_1 FLOPs for one forward pass through the large model $\mathcal{M}(\boldsymbol{\theta})$ and C_2 for proxy model $\mathcal{M}_{proxy}(\boldsymbol{\theta}')$. To train the model for T steps on $T \cdot |b_t|$ samples, RHO-LOSS takes $\mathcal{O}(TC_1 \cdot (|B_t| + |b_t|) + TC_2|B_t|)$ forward FLOPs. In case that $|B_t| \gg |b_t|$, the extra computation overhead would be significant.

3 Irreducible Curriculum

Aiming to reduce the computation cost while improving the training efficiency and robustness, we tend to design a pre-defined curriculum to simulate the online assessment of the scores $L[y|x;\mathcal{D}_{\mathrm{t}}]$ along the training process. We refer to our algorithm as IRREDUCIBLE CURRICULUM, since it is determined only by a proxy-model trained on a holdout dataset without resorting any large model's training dynamics.

Specifically, we train a small proxy model $\mathcal{M}_{proxy}(\boldsymbol{\theta'})$ on holdout set $\mathcal{D}_{\mathrm{ho}}$ for T overall steps while saving an early-stage checkpoint at t_0 steps ($\mathcal{M}_{proxy}(\boldsymbol{\theta'})[t_0]$) and a late-stage checkpoint($\mathcal{M}_{proxy}(\boldsymbol{\theta'})[T]$). We define the *learnability score* for every sample $(x,y)\in\mathcal{D}_{\mathrm{t}}$ of the training set as the gap between the loss from early-stage and the late-stage proxy model checkpoints[1]:

$$Learnability(x) := L_{early} - L_{late} = L\left[x;\boldsymbol{\theta'}[t_0]\right] - L\left[x;\boldsymbol{\theta'}[T]\right] \tag{3}$$

To differentiate better among data points, we take the average of loss from last three checkpoints as the late-stage loss.

The initial threshold of learnability is set as the top-$\lambda_0\%$ instance's score among the training set. The first training batch would be uniformly selected from these top-$\lambda_0\%$ instances whose learnability scores are above the threshold. To simulate the decreasing training loss in online data selection (Equ. 2), the threshold of learnability is decreasing along the training process following a step function. Specifically, consider that we are applying the curriculum on model $\mathcal{M}(\boldsymbol{\theta})$ during the first T_c steps (train for T steps in total), at step $t<T_c$, the training batch for $\mathcal{M}(\boldsymbol{\theta})$ is uniformly sampled from $(\lambda_0+(1-\lambda_0)t/T_c)$ instances from the training set with highest learnability. Thus, the sampling probability distribution across training set $\mathcal{D}_{\mathrm{t}}$ is:

$$\begin{aligned}&\mathcal{P}(t) = Unif\left(\{learnability(x) > S(t), x\in\mathcal{D}_{\mathrm{t}}\}\right), \\ &\textit{where } S(t) \textit{ satisfies } \frac{|\mathcal{D}_{\mathrm{t}} > S(t)|}{|\mathcal{D}_{\mathrm{t}}|} = \lambda_0 + (1-\lambda_0)t/T_c.\end{aligned} \tag{4}$$

After T_c steps, the uniform random sampling would be applied on the whole dataset. Our proposed IRREDUCIBLE CURRICULUM introduces no extra forward passes through the large language model $\mathcal{M}(\boldsymbol{\theta})$ when we pre-compute the learnability score using the proxy model. Instead, the only extra overhead comes from the proxy model as $\mathcal{O}(C_2\cdot(|D_t|))\geq\mathcal{O}(T_cC_2\cdot(|b_t|))$ forward FLOPs, which could be negligible comparing with $\mathcal{O}(TC_1\cdot(|B_t|+|b_t|)+TC_2|B_t|)$ overhead from RHO-LOSS[2].

4 Experiments and Results

In this section, we present our initial inquiry experimental results on RedPajama-1B dataset, including 7 domains[3]. We randomly split each domain dataset into two subsets $\mathcal{D}_{proxy}$ and $\mathcal{D}_{train}$. $\mathcal{D}_{proxy}$ is used to train the proxy model, which is applied to score $\mathcal{D}_{train}$ and construct the curriculum. Considering a severe unbalance in domain scale, we uniformly sample from each domain, while applying our proposed IRREDUCIBLE CURRICULUM inside each domain. We also provide the results applying the curriculum on the whole dataset in Section 5.

We use a 6-layer (82M) model as the proxy model $\mathcal{M}_{proxy}(\boldsymbol{\theta'})$ to construct the curriculum, which is applied to train a 12-layer (124M) model $\mathcal{M}(\boldsymbol{\theta})$. Both the proxy model and the full size model follow the decoder-only transformer architecture [Vaswani et al., 2023] and share the same set of training hyperparameters (batch size 64, context length 512, learning rate 3e-4). We compare IRREDUCIBLE CURRICULUM (CURRICULUM) with random uniformly sampling baseline (RS) and anti-curriculum (ANTI-CURRICULUM), where we start with a fraction of λ_0 samples with lowest score, and then increase the sampling upper bound with the same rate.

4.1 Result and Analysis

[1] we omit writing y here, as for language modelling the labels y refers to next-token prediction on the current sequence x.

[2] C_1, C_2, B_t, b_t notations are kept consistent with Sec. 2.1

[3] Arxiv, Wikipedia, Book, Common-Crawl, C4, Stackexchange, Github.